

Table 1. **The broader robot-learning landscape: 225 representative works, compared on six axes.** Beyond the 77 systems placed in the taxonomy (Figure ??, Tables ??-??), this table catalogues the wider field, grouped into eleven areas and sub-grouped by family. Columns: **Year**, **Ships** (what the method contributes), **Learn** (primary learning signal: IL, RL, offline-RL, LLM, self-sup., model-based), **Eval** (real/sim/game/offline), **Embod.** (embodiment), and run-time **S.I.** (self-improvement: ✓ yes, ~ partial, ✗ no). A -- marks not-applicable or unverified. Every entry is a real, individually citeable paper.

System / work	Year	Ships	Learn	Eval	Embod.	S.I.
End-to-end & generalist policies (28)						
<i>Vision-language-action</i>						
3D-VLA [221]	2024	world-model	LLM	sim	arm	✗
ECoT [213]	2024	policy	IL	real	arm	✗
LLaRA [109]	2024	policy	LLM	sim	arm	✗
ManipLLM [110]	2024	policy	LLM	sim	arm	✗
RoboMamba [115]	2024	policy	LLM	sim	arm	✗
RoboPoint [212]	2024	plan	LLM	real	multi	✗
TinyVLA [193]	2024	policy	IL	real	arm	✗
TraceVLA [222]	2024	policy	IL	sim	arm	✗
DexVLA [194]	2025	policy	IL	real	multi	✗
FAST [152]	2025	repr	IL	real	arm	✗
OpenVLA-OFT [96]	2025	policy	IL	sim	arm	✗
<i>Generalist / multi-task</i>						
QT-Opt [90]	2018	policy	RL	real	arm	~
MT-Opt [91]	2021	policy	RL	real	arm	~
Gato [156]	2022	policy	IL	sim	multi	✗
PaLM-E [34]	2023	plan	LLM	real	multi	✗
Q-Transformer [17]	2023	policy	offline-RL	real	arm	✗
UniPi [35]	2023	policy	model-based	sim	arm	✗
VIMA [87]	2023	policy	IL	sim	arm	✗
GR-2 [16]	2024	policy	IL	real	arm	✗
LEO [77]	2024	policy	LLM	sim	multi	✗
Magma [202]	2025	weights	LLM	sim	arm	✗
<i>Video / action pretraining</i>						
ATM [192]	2024	policy	self-sup	sim	arm	✗
GR-1 [195]	2024	policy	IL	sim	arm	✗
HPT [186]	2024	repr	IL	sim	multi	✗
VPP [75]	2024	policy	IL	sim	arm	✗
Cosmos [146]	2025	world-model	self-sup	--	--	✗
LAPA [208]	2025	policy	self-sup	sim	arm	✗
Seer [180]	2025	policy	IL	sim	arm	✗
Imitation & diffusion policies (21)						
<i>Imitation learning</i>						
BC-Z [83]	2021	policy	IL	real	arm	✗
Implicit BC [41]	2021	policy	IL	sim	arm	✗
robomimic [128]	2021	bench	IL	sim	arm	✗
PerAct [176]	2022	policy	IL	sim	arm	✗
ACT/ALOHA [220]	2023	policy	IL	real	arm	✗
MimicGen [129]	2023	data	IL	sim	arm	✗
RVT [51]	2023	policy	IL	sim	arm	✗
Mobile ALOHA [46]	2024	policy	IL	real	mobile	✗
RoboAgent [11]	2024	policy	IL	real	arm	✗
RVT-2 [52]	2024	policy	IL	sim	arm	✗
UMI [29]	2024	data	IL	real	arm	✗
<i>Diffusion policies</i>						
Diffuser [84]	2022	plan	model-based	offline	--	✗
BESO [159]	2023	policy	IL	sim	arm	✗
Decision Diffuser [5]	2023	policy	offline-RL	offline	--	✗
Diffusion Policy [28]	2023	policy	IL	sim	arm	✗
3D Diffuser Actor [94]	2024	policy	IL	sim	arm	✗
DP3 [214]	2024	policy	IL	sim	arm	✗
Equivariant Diffusion Policy [185]	2024	policy	IL	sim	arm	✗

Table 1 (continued)

System / work	Year	Ships	Learn	Eval	Embod.	S.I.
MDT [158]	2024	policy	IL	sim	arm	✗
RDT-1B [117]	2024	weights	IL	real	arm	✗
SuSIE [12]	2024	policy	IL	sim	arm	✗
LLM/VLM planning & task-and-motion (30)						
<i>LLM task planning</i>						
COWP [32]	2022	plan	LLM	sim	mobile	~
ReAct [206]	2022	plan	LLM	game	--	~
SayCan [4]	2022	plan	LLM	real	mobile	✗
Socratic Models [216]	2022	plan	LLM	real	arm	✗
Zero-Shot Planners [78]	2022	plan	LLM	sim	--	✗
DEPS [190]	2023	plan	LLM	game	--	~
GLAM [15]	2023	policy	RL	game	--	✗
Grounded Decoding [80]	2023	plan	LLM	real	mobile	✗
ISR-LLM [224]	2023	plan	LLM	sim	--	~
ITP [108]	2023	plan	LLM	real	arm	~
KnowNo [157]	2023	plan	LLM	real	arm	✗
LLM-Planner [177]	2023	plan	LLM	sim	mobile	~
Reflexion [174]	2023	plan	LLM	game	--	✓
RoCo [127]	2023	plan	LLM	sim	multi	~
SayPlan [155]	2023	plan	LLM	real	mobile	~
Self-Refine [122]	2023	plan	LLM	--	--	✓
SMART-LLM [92]	2023	plan	LLM	sim	multi	✗
Tree-Planner [74]	2023	plan	LLM	sim	--	~
ReAd [219]	2024	plan	LLM	sim	multi	~
<i>LLM task-and-motion</i>						
AutoTAMP [23]	2023	plan	LLM	sim	--	~
Lang2LTL [116]	2023	plan	LLM	real	mobile	✗
LLM+P [111]	2023	plan	LLM	--	--	✗
LLM-GROP [33]	2023	plan	LLM	sim	mobile	✗
NL2TL [24]	2023	plan	LLM	offline	--	✗
DELTA [118]	2024	plan	LLM	sim	mobile	✗
LLM3 [188]	2024	plan	LLM	sim	arm	~
<i>Code / prompt as policy</i>						
LM Traj Generators [103]	2023	plan	LLM	real	arm	~
MOKA [112]	2024	plan	LLM	real	arm	✗
PIVOT [145]	2024	plan	LLM	real	mobile	~
ReKep [79]	2024	code	LLM	real	arm	~
LLM reward synthesis & data generation (17)						
<i>LLM reward synthesis</i>						
ELLM [36]	2023	reward	RL	game	--	✗
LAMP [2]	2023	reward	RL	sim	arm	✗
Motif [97]	2023	reward	RL	game	--	✗
Read and Reap [198]	2023	reward	RL	game	--	✗
VLM-RM [161]	2023	reward	RL	sim	humanoid	✗
Code as Reward [182]	2024	reward	RL	sim	--	✗
RL-VLM-F [189]	2024	reward	RL	sim	arm	✗
<i>Generative data / augmentation</i>						
CACTI [126]	2022	data	IL	real	arm	✗
DIAL [200]	2022	data	IL	real	mobile	✗
GenAug [25]	2023	data	IL	real	arm	✗
GenSim [187]	2023	data	IL	sim	arm	✗
ROSIE [211]	2023	data	IL	real	mobile	✗
Scaling Up Distilling Down [59]	2023	policy	IL	real	arm	✗
BBSEA [205]	2024	skill	IL	sim	arm	~
DexMimicGen [88]	2024	data	IL	sim	humanoid	✗
GenSim2 [76]	2024	data	IL	sim	arm	✗
RoboDreamer [223]	2024	world-model	model-based	--	arm	✗
Unsupervised skill discovery & hierarchical RL (25)						
<i>Unsupervised skill discovery</i>						
VIC [54]	2016	skill	RL	sim	--	✗

Table 1 (continued)

System / work	Year	Ships	Learn	Eval	Embod.	S.I.
Eigenoption [121]	2018	skill	RL	game	--	✗
VALOR [1]	2018	skill	RL	sim	--	✗
EDL [14]	2020	skill	RL	sim	--	✗
VISR [68]	2020	skill	RL	game	--	~
APS [113]	2021	skill	RL	game	--	✗
APT [114]	2021	skill	RL	game	--	✗
ProtoRL [207]	2021	repr	self-sup	sim	--	✗
WURL [69]	2022	skill	RL	sim	--	✗
BeCL [204]	2023	skill	RL	sim	--	✗
Choreographer [131]	2023	skill	model-based	sim	--	✗
CSD [149]	2023	skill	RL	sim	multi	✗
DGPO [22]	2024	skill	RL	sim	--	✗
<i>Hierarchical / option RL</i>						
h-DQN [100]	2016	policy	RL	game	--	✗
FeUdal Networks [183]	2017	policy	RL	game	--	✗
Option-Critic [9]	2017	skill	RL	game	--	✗
SNN4HRL [42]	2017	skill	RL	sim	multi	✗
HIRO [140]	2018	policy	RL	sim	legged	✗
MLSH [43]	2018	skill	RL	sim	multi	~
HAC [107]	2019	policy	RL	sim	arm	✗
MCP [151]	2019	skill	RL	sim	humanoid	✗
NPMP [134]	2019	skill	IL	sim	humanoid	✗
Relay Policy Learning [55]	2019	policy	IL	sim	arm	✗
HIDIO [217]	2021	skill	RL	sim	multi	✗
Director [60]	2022	policy	model-based	sim	--	✗
World models & model-based RL (24)						
<i>World models</i>						
World Models [58]	2018	world-model	model-based	game	--	✗
PlaNet [62]	2019	world-model	model-based	sim	--	✗
Dreamer [61]	2020	world-model	model-based	sim	--	✗
DreamerV2 [63]	2021	world-model	model-based	game	--	✗
DayDreamer [196]	2022	world-model	model-based	real	multi	✓
Iso-Dream [147]	2022	world-model	model-based	sim	--	✗
MWM [166]	2022	world-model	model-based	sim	arm	✗
TransDreamer [19]	2022	world-model	model-based	game	--	✗
DreamerV3 [64]	2023	world-model	model-based	game	--	✗
IRIS [135]	2023	world-model	model-based	game	--	✗
STORM [218]	2023	world-model	model-based	game	--	✗
SWIM [133]	2023	world-model	model-based	real	arm	✗
TWM [160]	2023	world-model	model-based	game	--	✗
DIAMOND [7]	2024	world-model	model-based	game	--	✗
Genie [13]	2024	world-model	self-sup	game	--	✗
UniSim [203]	2024	world-model	self-sup	real	arm	✗
<i>Model-based RL</i>						
PETS [30]	2018	world-model	model-based	sim	--	✗
MBPO [85]	2019	policy	model-based	sim	--	✗
MuZero [165]	2020	world-model	model-based	game	--	✗
SimPLe [89]	2020	world-model	model-based	game	--	✗
SLAC [105]	2020	repr	RL	sim	--	✗
EfficientZero [209]	2021	world-model	model-based	game	--	✗
TD-MPC [67]	2022	world-model	model-based	sim	--	✗
TD-MPC2 [66]	2024	world-model	model-based	sim	multi	✗
Representation learning & offline RL (17)						
<i>Visual representation learning</i>						
MVP [201]	2022	repr	self-sup	sim	arm	✗
PVR [148]	2022	repr	self-sup	sim	multi	✗
R3M [142]	2022	repr	self-sup	sim	arm	✗
LIV [119]	2023	reward	self-sup	real	arm	✗
VC-1 [124]	2023	repr	self-sup	sim	multi	✗
VIP [120]	2023	reward	self-sup	real	arm	✗
Voltron [93]	2023	repr	self-sup	sim	arm	✗

Table 1 (continued)

System / work	Year	Ships	Learn	Eval	Embod.	S.I.
DIGIT [104]	2020	--	--	real	hand	✗
Where2Act [138]	2021	repr	self-sup	sim	arm	✗
FlowBot3D [37]	2022	policy	self-sup	sim	arm	✗
HULC [132]	2022	policy	IL	sim	arm	✗
VAT-Mart [197]	2022	plan	RL	sim	arm	✗
See-to-Touch [56]	2023	policy	RL	real	hand	~
T-DEX [57]	2023	repr	self-sup	real	hand	✗
VRB [10]	2023	repr	self-sup	real	arm	✗
Datasets, benchmarks & simulators (16)						
<i>Datasets</i>						
RoboNet [31]	2019	data	model-based	real	arm	✗
Ego4D [53]	2022	data	--	offline	--	✗
BridgeData V2 [184]	2023	data	IL	real	arm	✗
Dobbe [167]	2023	data	IL	real	mobile	✗
RH20T [38]	2023	data	IL	real	arm	✗
DROID [95]	2024	data	IL	real	arm	✗
<i>Benchmarks</i>						
D4RL [44]	2020	bench	offline-RL	offline	--	✗
ManiSkill [139]	2021	bench	IL	sim	arm	✗
<i>Simulators</i>						
AI2-THOR [98]	2017	code	--	sim	mobile	✗
Habitat [164]	2019	code	--	sim	mobile	✗
SAPIEN [199]	2020	code	--	sim	arm	✗
ThreeDWorld [50]	2020	code	--	sim	multi	✗
iGibson [173]	2021	code	--	sim	mobile	✗
Isaac Gym [125]	2021	code	--	sim	multi	✗
Isaac Lab Orbit [137]	2023	code	--	sim	multi	✗
RoboCasa [144]	2024	bench	IL	sim	mobile	✗

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